Notes: Jayaratne and Strahan (QJE, 1996)

The Finance-Growth Nexus: Evidence from Bank Branch Deregulation

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1 Big picture

- **Research question.** Do financial markets directly affect real economic growth? More specifically, did the relaxation of intrastate bank branching restrictions in the United States cause states to grow faster?
- **Main idea.** Branching restrictions protected small local banks and limited geographic expansion. When states relaxed these restrictions, banks could consolidate, branch more freely, and face more competition. The paper asks whether this financial-sector reform raised state-level growth.
- **Empirical setting.** The paper uses the staggered timing of intrastate branch deregulation across U.S. states from the 1970s through the early 1990s. This creates a within-country panel setting with treated states, untreated states, state fixed effects, and year fixed effects.
- **Main empirical object.** The dependent variable is the annual real per capita growth rate of state income or output. The key explanatory variable is an indicator equal to one after a state allows intrastate branching through mergers and acquisitions.
- **Main finding.** Real per capita growth increases significantly after intrastate branch deregulation. The estimates in the basic growth regressions range from about 0.5 to 1.2 percentage points per year.
- **Mechanism.** The growth increase does not appear to come mainly from a larger volume of bank lending or higher investment. Instead, the paper finds evidence that loan quality improves: nonperforming loans, charge-offs, and insider loans decline after deregulation.
- **Interpretation.** Bank deregulation seems to improve the efficiency of intermediation. Banks do not simply lend more; they lend better.
- **Economic takeaway.** Finance matters for growth not only because it may mobilize savings, but because it can allocate capital to better projects and discipline weak lending practices.

2 Incremental contribution of the paper

- **From cross-country correlation to quasi-experimental evidence.** Earlier finance-growth studies often compare countries with different institutions, laws, histories, and data quality. This paper uses U.S. states, which share a national macroeconomic environment, currency, federal law, and many data conventions.
- **A cleaner policy shock.** Intrastate branch reform is a concrete change in financial regulation. The staggered timing across states makes it possible to compare growth before and after deregulation while using other states as controls.
- **Generalized difference-in-differences design.** The empirical design includes state fixed effects and time fixed effects. This removes permanent differences across states and common national shocks.
- **Robustness to regional cycles.** Because many states deregulated in the same era, the paper also allows year effects to vary across broad U.S. regions, reducing the concern that regional business cycles drive the results.

- **Mechanism evidence inside the banking sector.** Unlike most cross-country work, the paper can directly examine banking outcomes after deregulation. It studies lending volume and loan quality using bank balance-sheet data.
- **Distinguishing quantity from quality of finance.** The paper shows that growth rises without strong evidence of higher loan growth or higher manufacturing investment. This supports the view that financial development improves the efficiency of investment.
- **Policy relevance.** The paper speaks directly to banking deregulation debates, including the question of whether restrictions on branching reduce economic performance.

3 Institutional background and data

3.1 Intrastate branching restrictions

Historically, U.S. banks faced restrictions on geographic expansion both across states and within states. Intrastate restrictions limited a bank's ability to operate branches throughout its home state. These rules protected incumbent local banks and fragmented the banking system.

- Before the 1970s, many states prohibited or sharply limited statewide branching.
- From the mid-1970s onward, many states relaxed these restrictions.
- Reform typically occurred in two steps: first, allowing multibank holding companies to consolidate subsidiaries into branches through mergers and acquisitions; second, allowing de novo branching statewide.
- The paper focuses on the dates when branching through mergers and acquisitions was allowed, because M&A reform and de novo reform are highly correlated.

Interpretation. The policy shock is not simply a change in bank geography. It changes competition, the market for bank control, the ability to consolidate weak banks, and the efficiency of bank intermediation.

3.2 Expected banking-market effects

Prior work cited by the paper finds that branch reform changes banking markets substantially.

- Large holding companies acquire small banks and convert subsidiaries into branches.
- Previously sheltered local markets face more entry and competition.
- Small banks lose market share.
- Larger banking organizations may diversify better and reduce redundant overhead.
- The threat of acquisition may discipline bank management.

Interpretation. Branch deregulation can improve the average quality of the banking sector through both selection and incentives. Weak banks lose protection, and surviving banks may operate more efficiently.

3.3 State growth data

The paper uses two measures of annual state economic activity.

- **Personal income.** Real per capita personal income growth is available from 1972 to 1992.
- **Gross state product.** Real per capita gross state product growth is available from 1978 to 1991.
- Personal income measures income of state residents; gross state product measures income generated by factors located in the state.
- Nominal personal income is deflated with the Consumer Price Index. Gross state product is deflated with industry-specific price deflators.
- Population data come from the U.S. Census Bureau.

Interpretation. Personal income and gross state product are not identical, so using both provides a useful robustness check. Personal income is more available over time, while gross state product is better insulated from relative-price changes.

3.4 Bank lending and loan-quality data

The mechanism tests use commercial bank data aggregated to the state level.

- Total loans are measured from commercial banks' year-end Reports of Condition from 1972 to 1992.
- Commercial loans are the sum of commercial and industrial loans plus commercial real estate loans, available from 1977 to 1992.
- Nonperforming loans are loans more than 90 days past due plus nonaccrual loans, available from 1982 to 1992.
- Net charge-offs are gross charge-offs less recoveries, available from 1976 to 1992.
- Insider loans are loans to executive officers and principal shareholders, available from 1983 to 1992.

Interpretation. Loan-volume data test whether deregulation increased the quantity of bank credit. Loan-quality data test whether deregulation improved screening, monitoring, and bank governance.

3.5 Sample choices

- Delaware is dropped because a 1982 tax incentive caused credit-card banks to enter and sharply changed the state's banking sector.
- The year in which a state deregulates is dropped from the regressions.
- Alaska and Hawaii are dropped in regional-effects models because the regional grouping is based on the lower 48 states.

- WLS specifications weight observations by the size of the state economy, because economic data for small states may contain more measurement error and be more sensitive to idiosyncratic industry shocks.

4 Methodology and empirical specifications

4.1 Baseline fixed-effects growth model

For state i in year t, the baseline model is

$$\frac{Y_{t,i}}{Y_{t-1,i}} = \alpha_t + \beta_i + \gamma D_{t,i} + \varepsilon_{t,i},$$

where $Y_{t,i}$ is real per capita income or output, and $D_{t,i}$ equals one when state i allows intrastate branching through mergers and acquisitions.

- β_i are state fixed effects.
- α_t are year fixed effects.
- γ measures the change in growth after branch deregulation.
- The model is a generalized difference-in-differences design.

Interpretation. The design compares a state's growth after deregulation to its own pre-deregulation growth, net of national shocks and permanent state differences.

4.2 Regional-cycle specification

To address the possibility that regional business cycles coincide with deregulation, the paper estimates

$$\frac{Y_{t,i}}{Y_{t-1,i}} = \alpha_{t,j} + \beta_i + \gamma D_{t,i} + \varepsilon_{t,i},$$

where j indexes broad U.S. regions.

- Region-year fixed effects allow the business cycle to vary by region.
- This is costly in degrees of freedom but useful as a robustness check.
- The central growth effect remains positive and significant in the main specifications.

4.3 Fiscal policy controls

The paper adds two state fiscal-policy variables:

- state capital expenditure divided by income or product;
- state tax receipts divided by income or product.

This tests whether branch deregulation coincided with other pro-growth policy changes. The estimated branching effect changes little.

4.4 Lagged dependent variables

A concern is that states may deregulate at the trough of a state-specific business cycle. To address this, the paper adds three lags of the dependent variable.

- The lagged-growth model absorbs local business-cycle dynamics.
- The branching coefficient remains positive.
- Statistical significance weakens, but the result remains broadly robust.

4.5 Growth dynamics

To distinguish short-run from longer-run effects, the paper splits the deregulation dummy into three windows:

- years 1 to 5 after deregulation;
- years 6 to 10 after deregulation;
- years 10 and beyond after deregulation.

The model drops state fixed effects so that early deregulators can contribute information about long-run effects. It controls for initial per capita income and lagged growth.

Interpretation. The estimates suggest that growth effects are large in the first ten years and may diminish after ten years, though the data cannot sharply pin down the long-run effect.

4.6 Loan growth and loan quality

The paper estimates the same type of fixed-effects model using state-level loan growth and state-level loan-quality measures as dependent variables.

- Loan-growth regressions test whether deregulation increased the quantity of credit.
- Loan-quality regressions test whether deregulation improved how banks selected and monitored borrowers.
- The loan-quality results are much stronger than the loan-volume results.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether state-level economic growth rises after a state relaxes intrastate branch restrictions, relative to states that have not changed branching rules and relative to the state's own pre-reform growth path.

The evidence is strongest as a within-country policy-based panel design. It is not a randomized experiment, but it is much less exposed to cross-country omitted-variable concerns than simple international finance-growth regressions.

5.2 Why intrastate reform is useful

The timing of branch deregulation varies substantially across states. This variation allows the paper to control for many confounders:

- fixed state characteristics, such as geography, industrial composition, and long-run tax or regulatory climate;
- nationwide shocks, such as recessions, inflation, or federal policy changes;
- regional business cycles in the regional-effects specification.

5.3 Reverse-causality concern

A central concern is that states may have deregulated because legislators expected future growth and wanted banks to finance a coming boom.

The paper argues against this interpretation in three ways:

- Figure I shows that many states did not experience higher growth after deregulation in absolute terms, and early deregulators still outperformed controls.
- Loan growth does not consistently increase after branch reform.
- Adding commercial and industrial loan growth to the growth regression does not reduce the estimated deregulation effect.

Interpretation. The evidence is more consistent with deregulation improving the banking system than with deregulation being a response to anticipated loan demand.

5.4 Quantity versus quality of finance

The paper's most important mechanism distinction is between:

- **more finance:** higher lending and higher investment; and
- **better finance:** better screening, monitoring, and allocation of bank credit.

The data support the second channel more strongly. Loan growth shows weak and inconsistent increases. Loan quality improves sharply.

5.5 Economic magnitude

The baseline growth estimates are large relative to average annual state growth rates. The paper argues that this magnitude is plausible because the estimates are based on annual data and are likely to reflect the years soon after reform. Improvements in the value and quality of existing bank loan portfolios can have large short-run effects on measured output.

5.6 Limitations

- The policy change is not randomized, so remaining omitted-variable concerns cannot be ruled out completely.
- The paper cannot fully separate the effects of M&A branching from de novo branching because the two reforms occur close together.
- Loan-quality measures come from bank balance sheets rather than direct borrower productivity.
- The growth effect may combine transitional gains with longer-run improvements; the data are limited for long-run dynamics.
- The mechanism evidence is suggestive rather than a full structural decomposition of bank selection, competition, governance, and borrower allocation.

6 Tables and figures

6.1 Table I: Timing of intrastate branch deregulation

Read:

- Table I reports when each state relaxed restrictions on intrastate branching through mergers and acquisitions and when it permitted de novo branching.
- A group of states, including Alaska, Arizona, California, Delaware, Idaho, Maryland, North Carolina, Nevada, Rhode Island, South Carolina, South Dakota, Vermont, and the District of Columbia, had already deregulated by 1972.
- Many states deregulated after 1972, especially during the 1980s.
- The only states listed as not deregulated are Arkansas, Iowa, and Minnesota.
- The paper uses the M&A deregulation date to construct the main treatment indicator.

Economic message: The table provides the treatment timing for the empirical design. The staggered adoption of branch reform is the source of identification.

TABLE I
DESCRIPTION OF CHANGES IN INTRASTATE BRANCHING RESTRICTIONS SINCE 1972

States deregulated by 1972	Year M&A branch restrictions lifted	Year branch restrictions lifted via de novo branching
Alaska		
Arizona		
California		
DC		
Delaware		
Idaho		
Maryland		
North Carolina		
Nevada		
Rhode Island		
South Carolina		
South Dakota		
Vermont		
States that deregulated after 1972		
Alabama	1981	1990
Colorado	1991	Still restricted
Connecticut	1980	1988
Florida	1988	1988
Georgia	1983	Still restricted
Hawaii	1986	1986
Illinois	1988	1993
Indiana	1989	1991
Kansas	1987	1990
Kentucky	1990	Still restricted
Louisiana	1988	1988
Massachusetts	1984	1984
Maine	1975	1975
Michigan	1987	1988
Missouri	1990	1990
Mississippi	1986	1989
Montana	1990	Still restricted
North Dakota	1987	Still restricted
Nebraska	1985	Still restricted
New Hampshire	1987	1987
New Jersey	1977	Still restricted
New Mexico	1991	1991
New York	1976	1976
Ohio	1979	1989
Oklahoma	1988	Still restricted
Oregon	1985	1985
Pennsylvania	1982	1990

Table I, part 1: Intrastate branching deregulation dates.

TABLE I
(CONTINUED)

States that deregulated after 1972	Year M&A branch restrictions lifted	Year branch restrictions lifted via de novo branching
Tennessee	1985	1990
Texas	1988	1988
Utah	1981	1981
Virginia	1978	1987
Washington	1985	1985
Wisconsin	1990	1990
West Virginia	1987	1987
Wyoming	1988	Still restricted
States that have not deregulated		
Arkansas		
Iowa		
Minnesota		

Table I, part 2: Continued deregulation dates and never-deregulated states.

6.2 Table II and Figure I: Basic growth effects and state-level evidence

Read Table II:

- Table II reports the baseline growth regressions.
- The estimated increase in personal-income growth ranges from 0.51 to 1.19 percentage points.
- The estimated increase in gross-state-product growth ranges from 0.69 to 1.08 percentage points.
- All eight branching coefficients are positive and statistically significant at the 5 percent level.
- The result holds in OLS and WLS, and with or without regional year effects.

Read Figure I:

- Figure I compares each deregulating state's post-reform change in mean growth with the corresponding control-state change.
- Most deregulating states perform better than their controls.
- The figure suggests that the result is not driven by one or two outliers.

- Some states have lower growth after deregulation in absolute terms, which weakens the idea that states deregulated simply in anticipation of growth booms.

Economic message: The core fact is that growth rises relative to the control group after intrastate branching is allowed.

TABLE II
GROWTH REGRESSIONS: BASIC MODELS

Growth based on personal income:	Estimated percentage point change in growth	Adjusted R^2 (number of observations)
1. Basic model, OLS	0.94*	49%
	(0.26)	(1015)
2. Basic model, WLS	1.19*	70%
	(0.24)	(1015)
3. Regional effects, OLS	0.51*	62%
	(0.23)	(974)
4. Regional effects, WLS	0.59*	78%
	(0.18)	(974)
Growth based on gross state product:		
5. Basic model, OLS	1.03*	43%
	(0.36)	(668)
6. Basic model, WLS	1.08*	65%
	(0.30)	(668)
7. Regional effects, OLS	0.69*	54%
	(0.33)	(641)
8. Regional effects, WLS	0.84*	72%
	(0.24)	(641)

This table presents estimates of the increase in growth following relaxation of intrastate branching restrictions. Table I presents the dates at which each state relaxed its restrictions on branching. We present a regression of real, per capita growth based on personal income or gross state product on time and state fixed effects, and an indicator variable equal to one for states with no restrictions on branching via M&A. We also estimate a regional effects model in which the time effects are interacted with a set of four regional indicators. Delaware is dropped from all regressions, while Alaska and Hawaii are dropped from the regional effects model. Also, the year in which each state deregulated was dropped. Growth data for personal income are from 1972–1992 and for state product from 1978–1991. Standard errors appear below coefficients in parentheses. An asterisk indicates statistical significance at the 5 percent level; reported standard errors are heteroskedasticity-consistent [White 1980]. Coefficients are multiplied by 100.

Table II: Basic growth regressions.

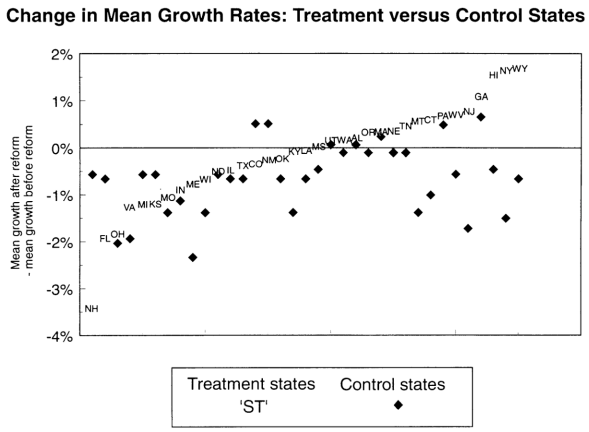

FIGURE I

Figure I: Treatment-state growth changes versus control-state changes.

6.3 Tables III and IV: Robustness to fiscal policy and business-cycle timing

Read Table III:

- Table III adds state capital expenditure and state tax receipts as controls.
- The branching coefficient remains positive and significant in all eight specifications.
- The estimates are very close to those in Table II.
- The result is therefore not explained by concurrent state fiscal policy changes.

Read Table IV:

- Table IV adds three lags of the dependent growth variable.
- This addresses the concern that states deregulated at the trough of a local business cycle.
- The branching coefficient remains positive in all eight specifications.
- Five estimates are significant at the 5 percent level, and two others are significant at the 10 percent level according to the paper's discussion.

Economic message: The growth result is robust to important confounds: state fiscal policy and local growth dynamics.

TABLE III

GROWTH REGRESSIONS: INCLUDING STATE FISCAL POLICY VARIABLES

Growth based on personal income:	Estimated percentage point change in growth	State capital expenditure/ income	State tax receipts/ income	Adjusted R^2 (number of observations)
1. Basic model, OLS	0.98*	0.012	−0.018	50%
	(0.25)	(0.019)	(0.014)	(994)
2. Basic model, WLS	1.20*	0.005	−0.012	71%
	(0.24)	(0.015)	(0.010)	(994)
3. Regional effects, OLS	0.61*	0.015	−0.013	63%
	(0.23)	(0.014)	(0.008)	(953)
4. Regional effects, WLS	0.63*	0.018	−0.006	78%
	(0.18)	(0.013)	(0.008)	(953)
Growth based on state product:				
5. Basic model, OLS	1.00*	−0.111*	0.028	44%
	(0.36)	(0.047)	(0.018)	(654)
6. Basic model, WLS	1.04*	−0.086*	0.029	65%
	(0.30)	(0.044)	(0.016)	(654)
7. Regional effects, OLS	0.70*	−0.107*	0.027	55%
	(0.33)	(0.041)	(0.014)	(627)
8. Regional effects, WLS	0.85*	−0.073	0.016	72%
	(0.25)	(0.039)	(0.013)	(627)

This table presents estimates of the increase in growth following relaxation of intrastate branching restrictions. Table I presents the dates at which each state relaxed its restrictions on branching. We present a regression of real, per capita growth based on personal income or gross state product on time and state fixed effects, an indicator variable equal to one for states with no restrictions on branching via M&A, the state government's capital expenditure per dollar of income (product), and the state government's tax receipts per dollar of income (product). We also estimate a regional effects model in which the time effects are interacted with a set of four regional indicators. Delaware and DC are dropped from all regressions, while Alaska and Hawaii are dropped from the regional effects model. Also, the year in which each state deregulated was dropped. Growth data for personal income are from 1972–1992 and for state product from 1978–1991. Standard errors appear below coefficients in parentheses. An asterisk indicates statistical significance at the 5 percent level; reported standard errors are heteroskedasticity-consistent [White 1980]. Coefficients are multiplied by 100.

Table III: Growth regressions with fiscal-policy variables.

TABLE IV

GROWTH MODEL WITH LAGGED DEPENDENT VARIABLES

Growth based on personal income:	Estimated percentage point change in growth	Growth_{t-1}	Growth_{t-2}	Growth_{t-3}	Adjusted R^2 (number of observations)
1. Basic model, OLS	0.88*	0.12	−0.05	−0.06	50%
	(0.27)	(0.09)	(0.06)	(0.08)	(1015)
2. Basic model, WLS	0.97*	0.18*	0.04	0.02	72%
	(0.23)	(0.05)	(0.04)	(0.04)	(1015)
3. Regional effects, OLS	0.49*	0.06	−0.04	0.01	62%
	(0.23)	(0.08)	(0.07)	(0.08)	(974)
4. Regional effects, WLS	0.52*	0.14*	0.02	0.05	78%
	(0.18)	(0.05)	(0.05)	(0.05)	(974)
Growth based on gross state product:					
5. Basic model, OLS	0.75	0.16*	0.05	−0.04	39%
	(0.45)	(0.06)	(0.05)	(0.08)	(521)
6. Basic model, WLS	0.85*	0.15*	0.08	−0.03	60%
	(0.37)	(0.05)	(0.06)	(0.07)	(521)
7. Regional effects, OLS	0.62	0.12*	0.03	0.02	48%
	(0.42)	(0.06)	(0.05)	(0.07)	(500)
8. Regional effects, WLS	0.52	0.18*	0.07	0.01	69%
	(0.30)	(0.05)	(0.05)	(0.07)	(500)

This table presents estimates of the increase in growth following relaxation of intrastate branching restrictions. Table I presents the dates at which each state relaxed its restrictions on branching. We present a regression of real, per capita growth based on personal income or gross state product on time and state fixed effects, three lags of the dependent variable, and an indicator variable equal to one for states with no restrictions on branching via M&A. We also estimate a regional effects model in which the time effects are interacted with a set of four regional indicators. Delaware is dropped from all regressions, while Alaska and Hawaii are dropped from the regional effects model. Also, the year in which each state deregulated was dropped. Growth data for personal income are from 1972–1992 and for state product from 1981–1991 (three years are lost with the addition of the lagged dependent variables). Standard errors appear below coefficients in parentheses. An asterisk indicates statistical significance at the 5 percent level; reported standard errors are heteroskedasticity-consistent [White 1980]. The coefficient on the branching indicator is multiplied by 100.

Table IV: Growth regressions with lagged dependent variables.

6.4 Table V: Growth dynamics after deregulation

Read:

- Table V separates the deregulation effect into years 1-5, years 6-10, and years 10 and beyond.
- Growth effects are generally positive during the first ten years after deregulation.
- The ten-year-plus coefficient is usually smaller than the first-ten-year coefficients.
- In personal-income specifications, the paper describes effects of roughly 0.3 to 0.9 percentage points during the first ten years, slowing to about 0.2 to 0.3 percentage points after ten years.
- The data cannot precisely determine whether the effect eventually disappears or remains positive in the long run.

Economic message: The growth acceleration is strongest soon after reform. Some longer-run effect may remain, but the paper is cautious about drawing sharp long-run conclusions.

TABLE V
GROWTH DYNAMICS

Growth based on personal income	Growth effect: Years 1–5	Growth effect: Years 6–10	Growth effect: Years 10+	Growth_{t-1}	Gr
1. Basic model, OLS	0.59*	0.86*	0.34	0.14	–
	(0.23)	(0.23)	(0.22)	(0.08)	
2. Basic model, WLS	0.61*	0.86*	0.34*	0.20*	
	(0.21)	(0.22)	(0.16)	(0.05)	
3. Regional effects, OLS	0.35	0.37	0.17	0.08	–
	(0.20)	(0.20)	(0.19)	(0.08)	
4. Regional effects, WLS	0.31*	0.38*	0.21	0.16*	
	(0.16)	(0.19)	(0.13)	(0.05)	
Growth based on gross state product					
5. Basic model, OLS	0.77*	0.94*	0.63*	0.21*	
	(0.30)	(0.30)	(0.27)	(0.06)	
6. Basic model, WLS	0.64*	0.83*	0.48	0.21*	
	(0.26)	(0.33)	(0.26)	(0.05)	
7. Regional effects, OLS	0.60*	0.65*	0.67*	0.15*	
	(0.29)	(0.27)	(0.27)	(0.06)	
8. Regional effects, WLS	0.43*	0.57*	0.59*	0.23*	
	(0.21)	(0.24)	(0.24)	(0.04)	

This table presents estimates of the increase in growth following relaxation of intrastate branching restric
restrictions on branching. We present a regression of real, per capita growth based on personal income or gross s
$t-1$, three lags of the dependent variable, and three nonoverlapping indicator variables for states with no restricti
one for years 6 to 10 after deregulation, and one for all years beyond 10. We also estimate a regional effects model
indicators. Delaware is dropped from all regressions, while Alaska and Hawaii are dropped from the regional e
dropped. Growth data for personal income are from 1972–1992 and for state product from 1981–1991 (three years
errors appear below coefficients in parentheses. An asterisk indicates statistical significance at the 5 percent level;
1980). The coefficients on the branching indicators and the lag of income are multiplied by 100.

Table V, part 1: Growth effects by years since deregulation.

TABLE V
GROWTH DYNAMICS

Growth based on personal income	Growth effect: ars 10+	Growth_{t-1}	Growth_{t-2}	Growth_{t-3}	Lag of per capita income	Adjusted R^2 (number of observations)
1. Basic model, OLS	0.34	0.14	−0.03	−0.04	−0.38*	52%
	(0.22)	(0.08)	(0.06)	(0.08)	(0.13)	(1015)
2. Basic model, WLS	0.34*	0.20*	0.06	0.04	−0.29*	73%
	(0.16)	(0.05)	(0.04)	(0.04)	(0.08)	(1015)
3. Regional effects, OLS	0.17	0.08	−0.03	0.02	−0.29*	64%
	(0.19)	(0.08)	(0.07)	(0.08)	(0.11)	(974)
4. Regional effects, WLS	0.21	0.16*	0.04	0.07	−0.28*	79%
	(0.13)	(0.05)	(0.04)	(0.05)	(0.09)	(974)
Growth based on gross state product						
5. Basic model, OLS	0.63*	0.21*	0.09	0.03	−0.07*	41%
	(0.27)	(0.06)	(0.05)	(0.07)	(0.03)	(521)
6. Basic model, WLS	0.48	0.21*	0.13*	0.06	−0.09*	62%
	(0.26)	(0.05)	(0.06)	(0.07)	(0.03)	(521)
7. Regional effects, OLS	0.67*	0.15*	0.06	0.07	−0.04	50%
	(0.27)	(0.06)	(0.05)	(0.07)	(0.02)	(500)
8. Regional effects, WLS	0.59*	0.23*	0.11*	0.08	−0.08*	69%
	(0.24)	(0.04)	(0.04)	(0.07)	(0.03)	(500)

This table presents estimat ation of intrastate branching restrictions. Table I presents the dates at which each state relaxed its
restrictions on branching. We pr based on personal income or gross state product on time fixed effects, state per capita income at time
$t-1$, three lags of the dependent r variables for states with no restrictions on branching via M&A: one for years 1 to 5 after deregulation,
one for years 6 to 10 after dereg also estimate a regional effects model in which the time effects are interacted with a set of four regional
indicators. Delaware is dropped waii are dropped from the regional effects model. Also, the year in which each state deregulated was
dropped. Growth data for person product from 1981–1991 (three years are lost with addition of the lagged dependent variables). Standard
errors appear below coefficients cal significance at the 5 percent level; reported standard errors are heteroskedasticity-consistent [White
1980]. The coefficients on the br multiplied by 100.

Table V, part 2: Lag controls, convergence control, and continued notes.

6.5 Tables VI and VII: Lending volume versus lending quality

Read Table VI:

- Table VI tests whether bank lending grows faster after intrastate branching reform.
- The evidence is weak and inconsistent.
- Only two of eight coefficients are statistically significant at the 5 percent level.
- Commercial-loan growth does not rise reliably.
- The paper also reports that controlling for commercial and industrial loan growth does not reduce the estimated growth effect.

Read Table VII:

- Table VII tests whether bank loan quality improves after deregulation.
- Nonperforming loans fall by 0.24 to 0.77 percentage points, relative to an unconditional mean of about 2 percent.
- Charge-offs fall significantly in all four specifications.
- Insider loans fall significantly in all four specifications.
- The paper interprets these results as evidence that screening, monitoring, or bank governance improves after branching restrictions are lifted.

Economic message: The key transmission channel is not more bank lending. It is better bank lending.

TABLE VI
BANK LOAN GROWTH FOLLOWING INTRASTATE BRANCH DEREGULATION

Real growth in loans (mean = 1.0%):	Estimated percentage point change in loan growth	Adjusted R^2 (number of observations)
1. Basic model, OLS	0.79	6%
	(3.60)	(1015)
2. Basic model, WLS	2.62*	21%
	(1.22)	(1015)
3. Regional effects, OLS	0.40	7%
	(2.78)	(974)
4. Regional effects, WLS	1.85*	23%
	(0.87)	(974)
Real growth in commercial loans (mean = 1.0%):		
5. Basic model, OLS	−0.47	4%
	(5.12)	(767)
6. Basic model, WLS	3.29	9%
	(1.70)	(767)
7. Regional effects, OLS	−1.20	5%
	(4.00)	(736)
8. Regional effects, WLS	2.34	9%
	(1.24)	(736)

This table presents estimates of the change in loan growth (aggregated to the state level) following relaxation of intrastate branching restrictions. Table I presents the dates at which each state relaxed its restrictions on branching. We present a regression of the growth rate of all loans and the growth rate of commercial loans (commercial and industrial loans plus commercial real estate loans) on state and time fixed effects and an indicator variable equal to one for states with no restrictions on branching via M&A. We also estimate a regional effects model in which the time effects are interacted with a set of four regional indicators. Delaware is dropped from all regressions, while Alaska and Hawaii are dropped from the regional effects model. Also, the year in which each state deregulated was dropped. Loan growth data are from 1972–1992; commercial loan data are from 1977–1992. Standard errors appear below coefficients in parentheses. An asterisk indicates statistical significance at the 5 percent level; reported standard errors are heteroskedasticity-consistent [White 1980]. Coefficients are multiplied by 100.

Table VI: Bank loan growth after intrastate branch deregulation.

TABLE VII
BANK LOAN QUALITY REGRESSIONS

Nonperforming loans/loans (mean = 2%):	Estimated percentage point change	Adjusted R^2 (number of observations)
1. Basic model, OLS	−0.77*	46%
	(0.17)	(523)
2. Basic model, WLS	−0.24	59%
	(0.19)	(523)
3. Regional effects, OLS	−0.63*	60%
	(0.15)	(502)
4. Regional effects, WLS	−0.34*	73%
	(0.12)	(502)
Charge-offs/loans (mean = 0.76%):		
5. Basic model, OLS	−0.61*	38%
	(0.08)	(816)
6. Basic model, WLS	−0.46*	47%
	(0.07)	(816)
7. Regional effects, OLS	−0.46*	52%
	(0.07)	(783)
8. Regional effects, WLS	−0.27*	61%
	(0.05)	(783)
Loans to insiders/loans (mean = 0.46%):		
9. Basic model, OLS	−0.15*	66%
	(0.05)	(474)
10. Basic model, WLS	−0.13*	59%
	(0.04)	(474)
11. Regional effects, OLS	−0.20*	68%
	(0.05)	(455)
12. Regional effects, WLS	−0.11*	63%
	(0.04)	(455)

This table presents estimates of the change in bank loan quality (aggregated to the state level) following relaxation of intrastate branching restrictions. Table I presents the dates at which each state relaxed its restrictions on branching. We estimate a regression of the loan quality variable on time and state fixed effects and an indicator variable equal to one for states with no restrictions on branching via M&A. We include three measures of loan quality: nonperforming loans to total loans (loans more than 90 days past due plus nonaccrual loans), net charge-offs (gross charge-offs minus recoveries) on loans to total loans, and loans to insiders to total loans (executives and principal shareholders). Nonperforming loans are from 1982–1992, charge-offs are from 1976–1992, and insider loans are from 1983–1992. Delaware is dropped from all regressions, while Alaska and Hawaii are dropped from the regional effects model. Also, the year in which each state deregulated was dropped. Standard errors appear below coefficients in parentheses. An asterisk indicates statistical significance at the 5 percent level; reported standard errors are heteroskedasticity-consistent [White 1980]. Coefficients are multiplied by 100.

Table VII: Bank loan quality after intrastate branch deregulation.

7 Short summary

- The paper studies whether financial-market reform can cause economic growth.
- The setting is staggered intrastate bank branching deregulation across U.S. states.
- The main empirical design is a state-year fixed-effects model, equivalent to a generalized difference-in-differences design.
- Growth is measured using real per capita personal income and real per capita gross state product.
- States grow faster after intrastate branching restrictions are relaxed.
- The basic estimates imply growth increases of roughly 0.5 to 1.2 percentage points per year.
- The result survives controls for regional cycles, fiscal policy, and lagged state growth.
- The paper finds little consistent evidence that total lending or commercial lending increases after reform.
- Loan quality improves substantially: nonperforming loans, charge-offs, and insider loans decline.

- The main conclusion is that finance affects growth primarily by improving the efficiency of capital allocation, not by increasing the sheer volume of investment.

8 Conclusion

8.1 Core conclusion

Jayaratne and Strahan show that U.S. states grow faster after they relax intrastate bank branching restrictions. The effect is economically large and robust across several specifications. The evidence supports the view that financial-sector reform can have real effects on economic performance.

8.2 Economic intuition

Branching restrictions protect local banks from competition and limit the ability of efficient banks to expand. Deregulation permits consolidation, competition, and more effective bank monitoring. If better banks allocate credit to better projects and reduce poor lending, then a state can grow faster even without a large increase in the volume of credit.

8.3 Contribution revisited

The paper makes the finance-growth relationship more concrete. Instead of relying on cross-country correlations between financial depth and growth, it studies a specific financial reform within one country and traces the mechanism to bank balance-sheet outcomes.

8.4 Implications

The findings imply that financial regulation affects real activity through the quality of intermediation. Policy debates about banking structure should therefore focus not only on bank safety or consumer access, but also on whether the financial system allocates capital efficiently.

8.5 Limitations

The evidence is not a randomized experiment. Some state-specific changes may still be correlated with deregulation, even though the paper performs many robustness checks. The loan-quality evidence is based on bank balance sheets, not direct borrower productivity. The paper also cannot perfectly separate the effects of different forms of branching reform.

8.6 Takeaway

The enduring takeaway is that finance can cause growth when it improves the allocation and monitoring of capital. In this setting, bank branch deregulation appears to make banks better lenders, and better lending is associated with faster state economic growth.